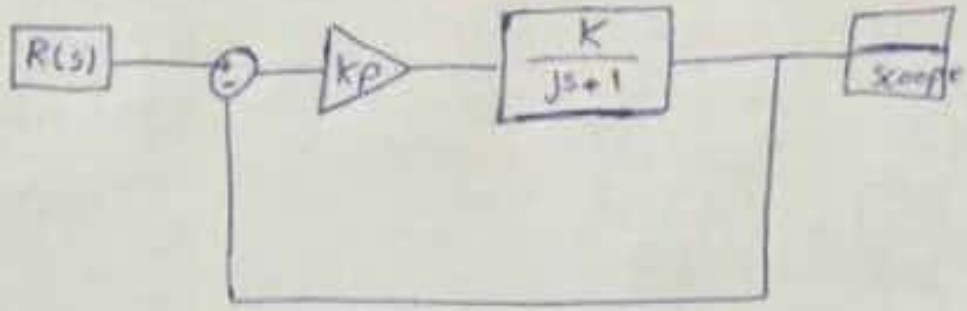
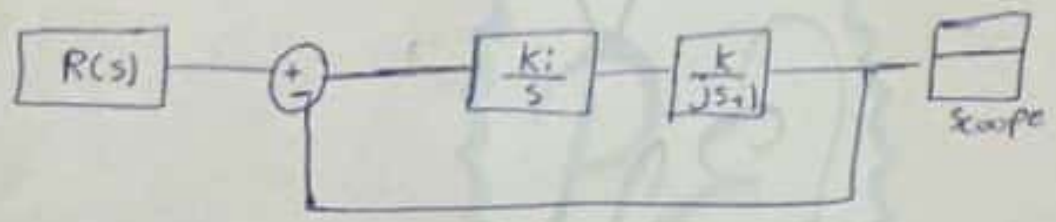
- second order system

A) varying kp in p controller



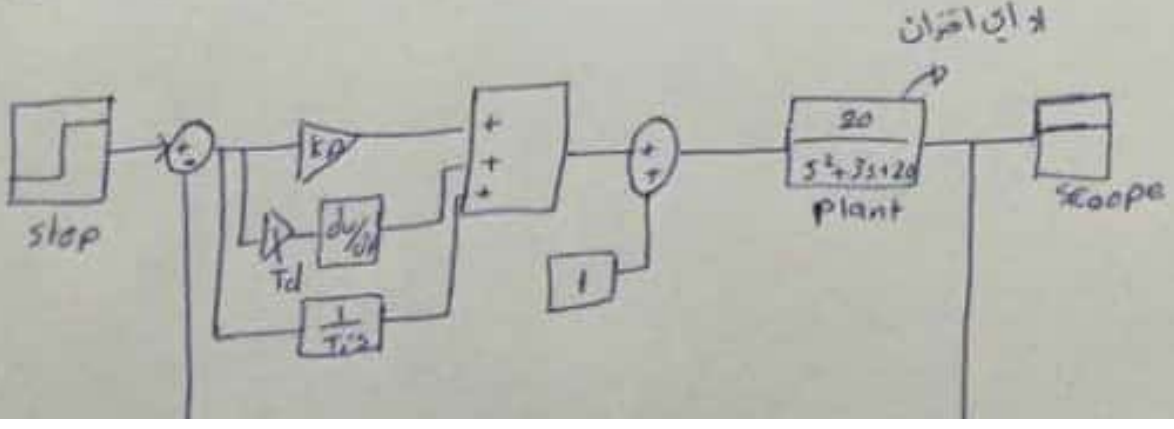
when increase kp↑ the over shoot increase
and Reduce steady state error

B) varying K_i in I controller



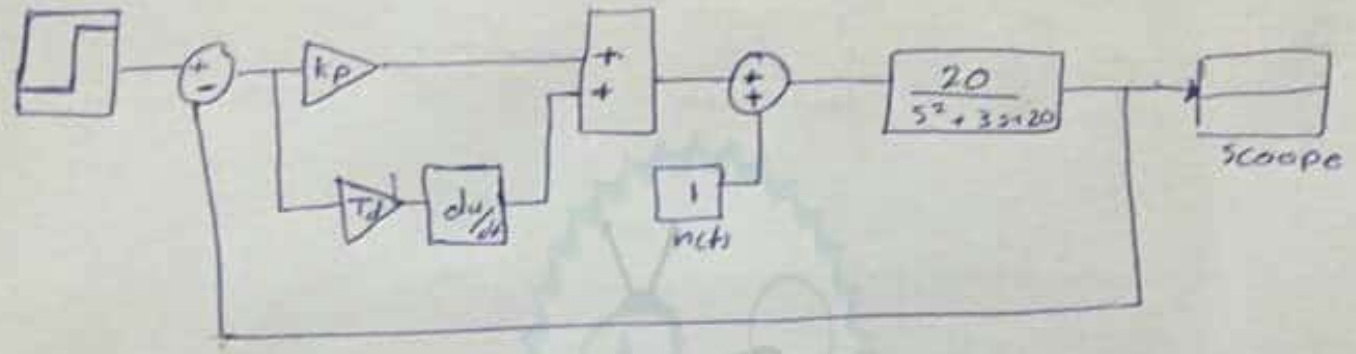
when K_i increase
the steady state error = zero

C) varying T_i in PID controller if kp = 5 and T_d = 10 to reach 20% of maximum over shoot



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D) varying Td in PD controller if $K_p = 10$



when Td increase in PD will reduce the oscillation and reduce the steady state error

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given $G(s) = \frac{1}{s(s+0.4)}$, negative unity feed back

unit step input, a) simulate this system

b) plot the output of the system

c) discuss the result ; d) select a controller

verify your selection

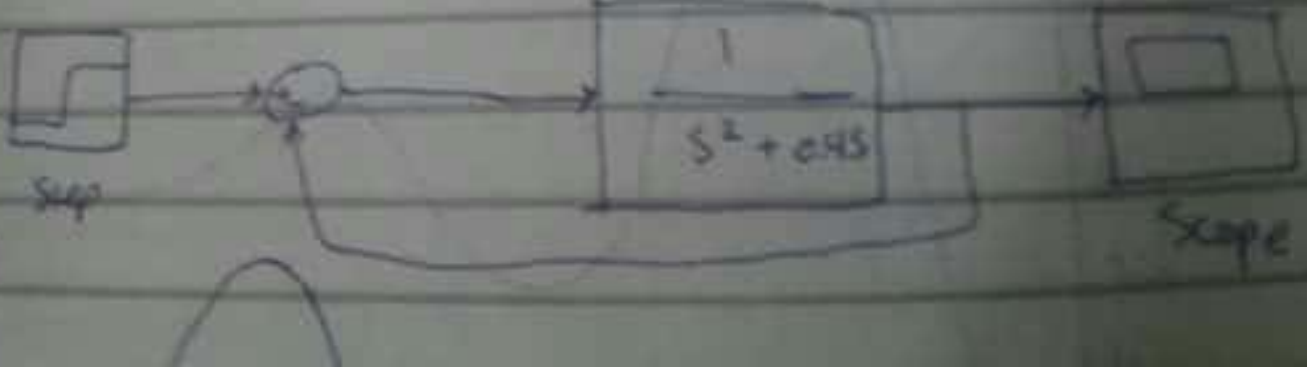
e) simulates the controlled system

f) plot and explain the response of controled syst

P 2 plot and explain the full load armature curre

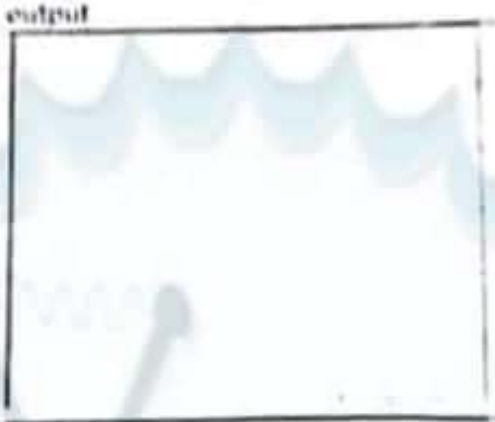
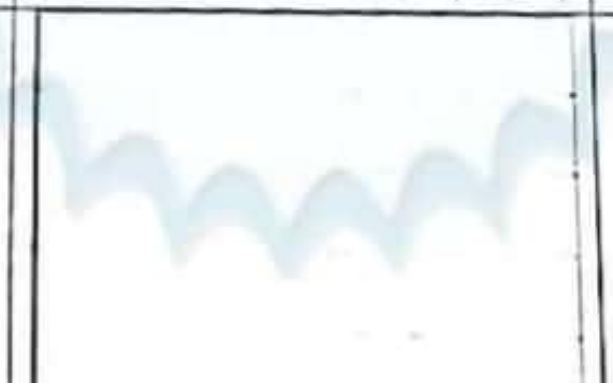
of a servo system.

Solution:-



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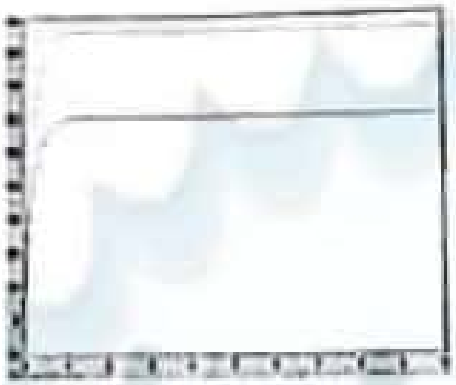
Q1:- [Marks] For second order system plot the simulation and output and step output

Circuit	output	Command and type of response
Varying damping ratio [0.5 / 1 / 4]		
Varying damping ratio [0 / -0.1]		
Varying Kp [2 , 6]		
If Kp=10 Varying Td [1 , 6]		

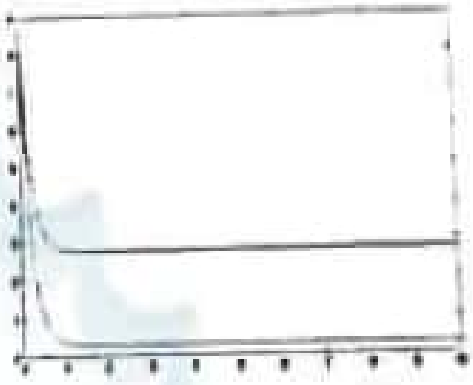
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Name

c. [10 marks] The following curves are the output of dc motor (open loop)



Speed vs Time



Current vs Time

a. Which plot represents the case of no load in each curve

b. Explain the effect of load on ia and w

c. Sketch ia and w in case of position closed loop system on the above case

Q3) [10 marks] a) Determine the state space matrices A,B,C & D for a system given

$$Y'''(t) + 4Y''(t) + 0.5Y' = 2r(t)$$

Parameters

A:

B:

C:

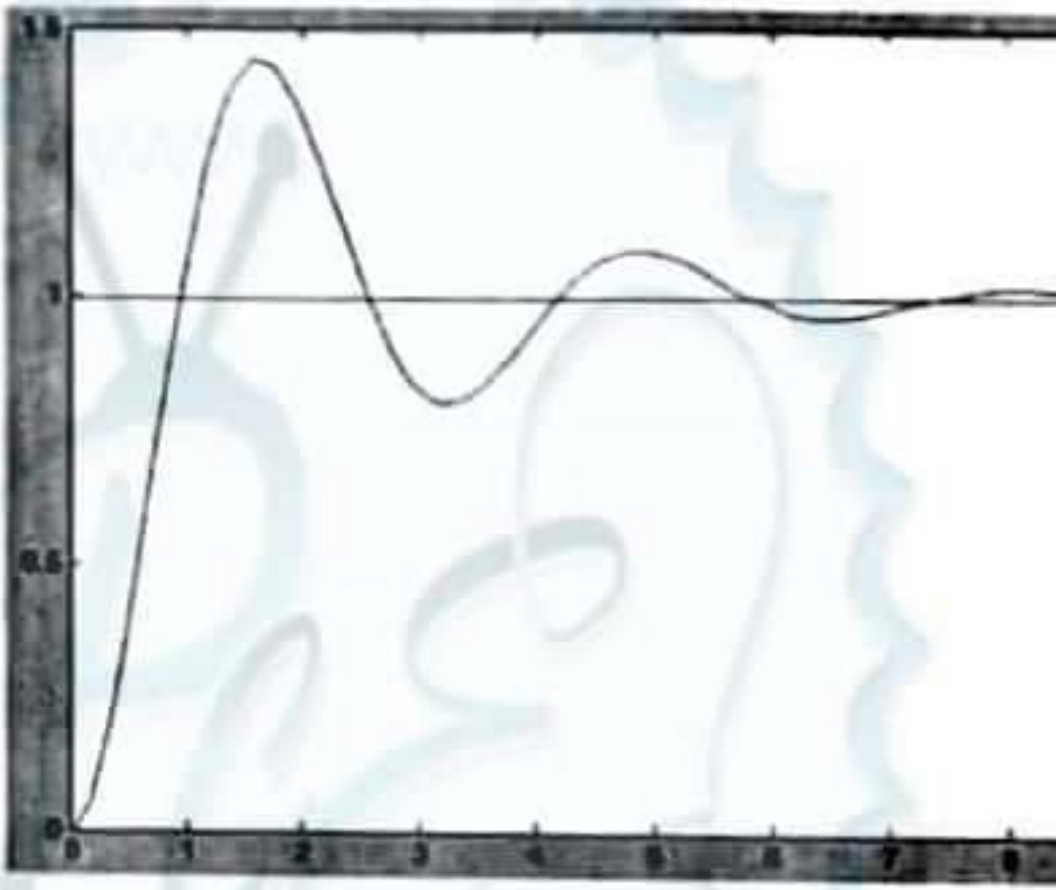
D:

b) Find the transfer function of above equation

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Q2) [10 marks] From the plotted curve obtain the time response specifications of the curve

td	tr	tp	Mp



Q3) [10 marks]

Explain

A. Increase proportional controller in first order system

when increase kP

B. Increase integral controller in second order system

C. Increase Time constant in first order system

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Q2) [10 marks] The unit step response of a second order system is shown for wn = 1 and ζ = 0.5

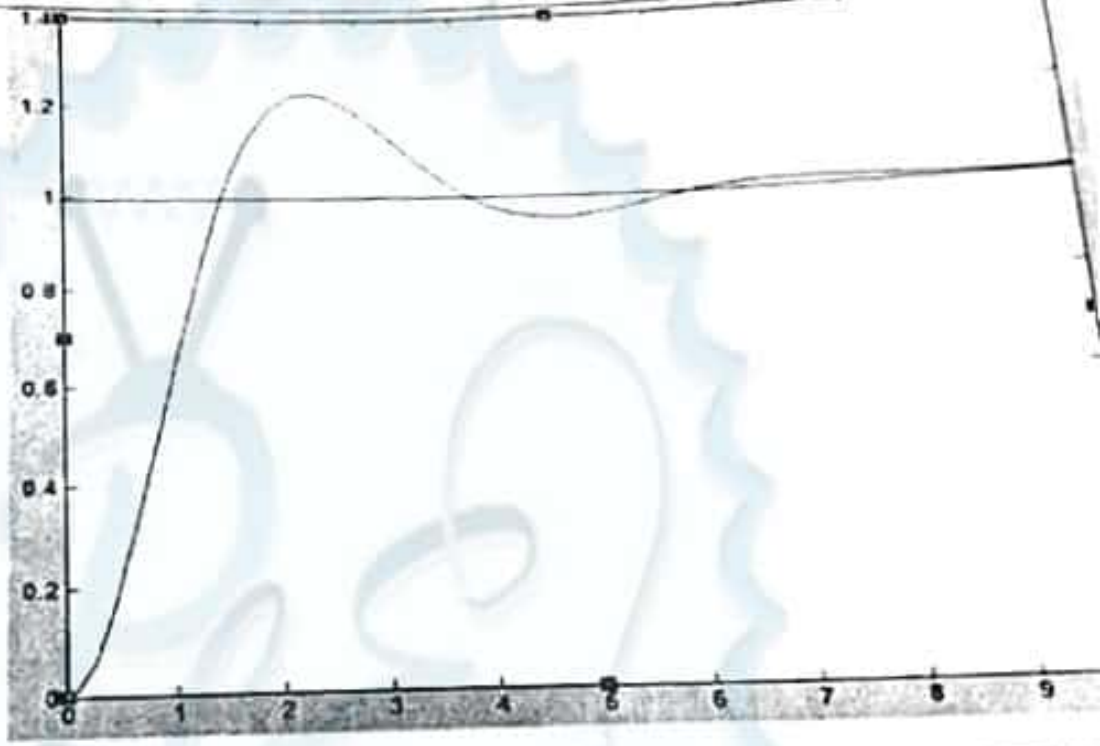
- On the same figure plot the response for

1) r(t) =1, wn =1, ζ=1

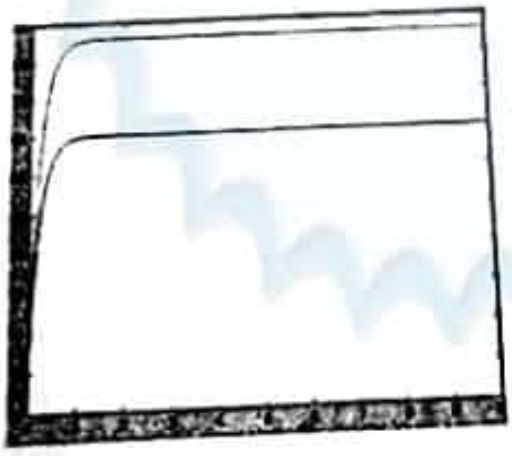
2) r(t) =1, wn =1, ζ=0

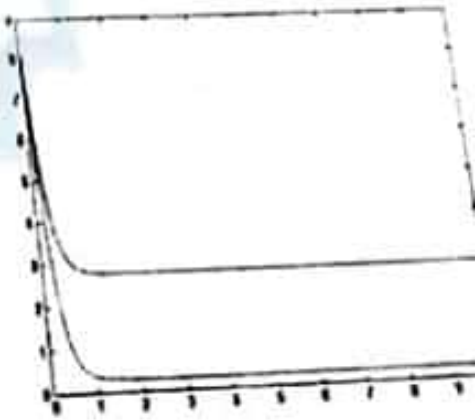
3) r(t) =2, wn =1, ζ=0.5

4) r(t) =1, wn =2, ζ=0.5



Q3) [10 marks] The following curves are the output of dc motor (open loop)





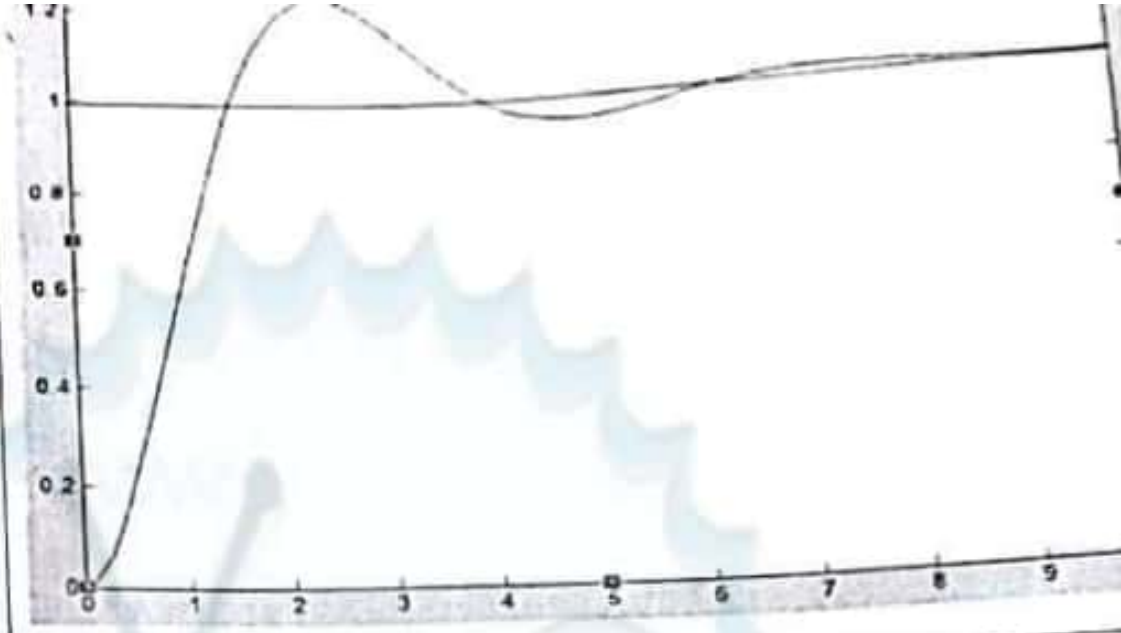
Current vs Time

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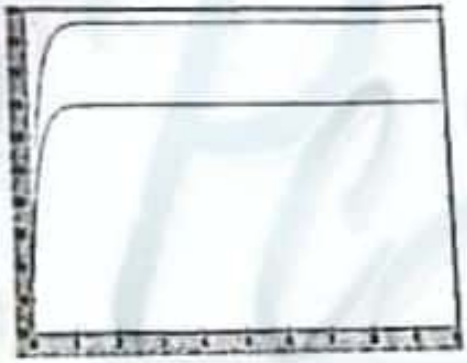
2) r(t) =1, wn =1, ζ=0

3) r(t) =2, wn =1,ζ=0.5

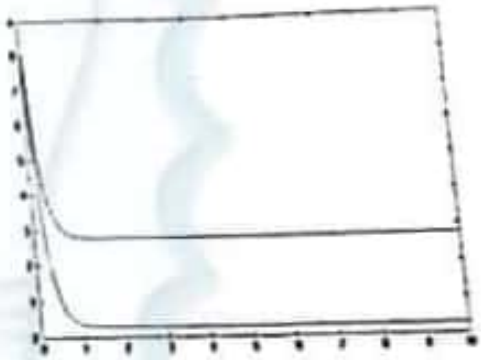
4) r(t) =1, wn =2, ζ=0.5



Q3) [10 marks] The following curves are the output of dc motor (open loop)



Speed vs Time



Current vs Time

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